Bell-State Preparation on a Two-Legged Ladder QPU

Dynamic swap, and unitary GHZ-uncompute

Overnight Research Company

2026 NCCU Applied Physics Challenge — 14 May 2026

The challenge

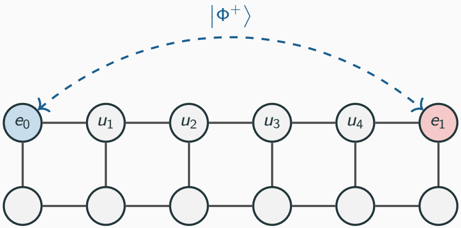
Goal. Prepare a Bell state

$$\left|\Phi^+\right\rangle_{e_0 e_1} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

between the two *far-end* qubits of a two-legged ladder QPU, for any ladder length L.

Constraints.

- Only nearest-neighbour gates (ladder edges).
- Top leg, bottom leg, rungs all allowed.
- Must scale to arbitrary L.



The challenge example: $L = 5$ (12 qubits).

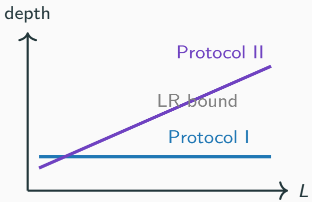
Why is this hard?

Lieb–Robinson bound
For any *unitary* circuit of nearest-neighbour gates on a 1D chain, information cannot travel faster than one site per layer:

$$\text{depth}(\text{unitary 1D protocol}) \geq \Omega(L).$$

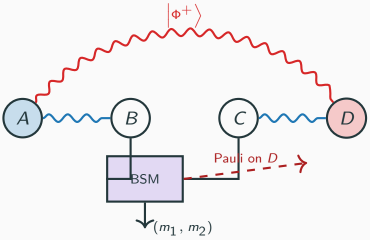
Two complementary ways out:

1. Break unitarity — use mid-circuit measurement + classical feed-forward. $\Rightarrow$ **Protocol I** ($O(1)$ depth).
2. Saturate the LR bound with a unitary depth-optimal construction. $\Rightarrow$ **Protocol II / MOGU** (depth $L+2$, 0 measurements).



Motivation A: entanglement swapping

Primitive (Żukowski–Zeilinger–Horne–Ekert, 1993): start with two Bell pairs (A, B), (C, D) — Bell-measure the middle pair (B, C) — Pauli-correct D based on the outcome — A, D end up in $|\Phi^+\rangle$ though they never interacted.

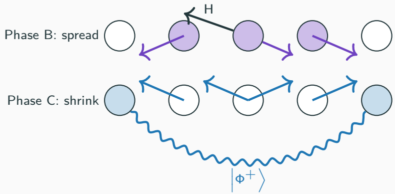


Bäumer *et al.*, PRX Quantum **5**, 030339 (2024) realised this primitive on 101 IBM qubits with constant-depth dynamic circuits.

Motivation B: GHZ-uncompute (the unitary alternative)

Idea. If you cannot measure, build the entanglement everywhere and then carefully *uncompute* what you don't want.

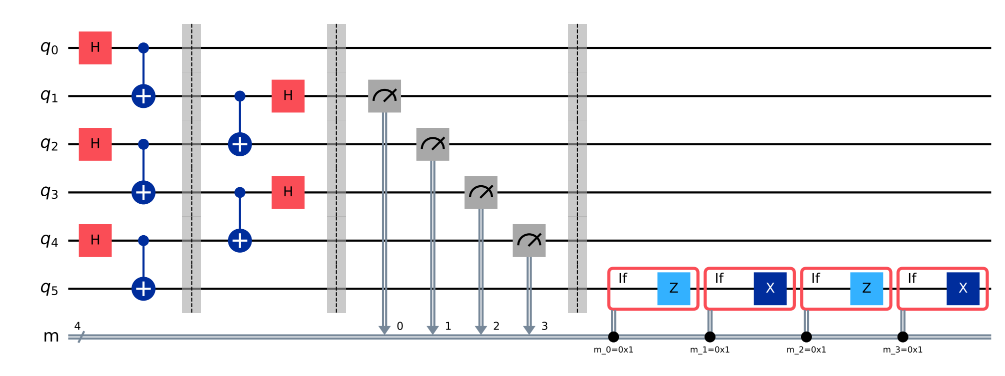
1. Hadamard a central qubit, fan CNOTs outward $\Rightarrow$ GHZ $\frac{1}{\sqrt{2}}(|0\cdots0\rangle + |1\cdots1\rangle)$ across the top leg.
2. Reverse the fan-out from the centre outward $\Rightarrow$ interior qubits return to $|0\rangle$; only the endpoints stay entangled.



Cost: depth $L+2$, $2L-1$ CNOTs, zero measurements. We call this **MOGU** = *Middle-Out GHZ-Uncompute*.

Protocol I — entanglement-swap with feed-forward

Three layers, $O(1)$ depth: (1) three Bell pairs on alternating links; (2) two parallel Bell-measurements on the inner pairs; (3) classically-controlled Pauli correction on e_1.



Protocol I — the core math

After the Bell-measurement layer, the residual stabilisers on (e_0, e_1) reduce to

$$\tilde{P}_{XX} = (-1)^{m_1+m_3}\, X_{e_0} X_{e_1}, \qquad \tilde{P}_{ZZ} = (-1)^{m_2+m_4}\, Z_{e_0} Z_{e_1}.$$

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Feed-forward Pauli correction on e_1:

$$\boxed{X^{m_2 \oplus m_4}\ Z^{m_1 \oplus m_3}}$$

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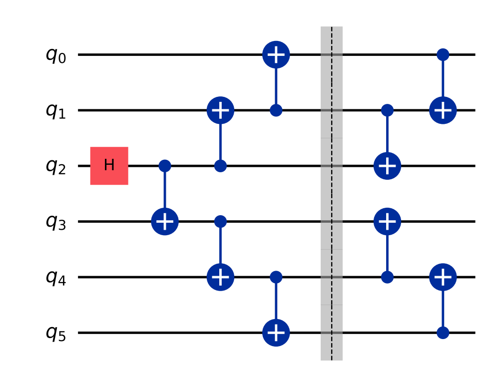
Generalisation

- Even N: induction on Bell-pair count $\Rightarrow$ depth ≤ 7 for all L.
- Odd N: a single GHZ-3 link at the tail handles the parity obstruction.

Resources. $N{+}1$ or $N{+}2$ two-qubit gates, N measurements, constant depth.

Protocol II — MOGU (measurement-free)

Three phases: (1) H on middle qubit $q_{\lfloor L/2 \rfloor}$; (2) spread CNOTs fan out from middle to ends; (3) shrink reverse-fan-out disentangles every interior qubit, centre outward.



Protocol II — core formulas

Sequential schedule.

$$\text{depth} = L + 2, \quad \text{2Q gates} = 2L - 1, \quad \text{measurements} = 0.$$

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Stabiliser sketch. After Phase B the state is the standard $(L{+}1)$-qubit GHZ. Phase C peels off each interior Z_i generator, leaving

$$\langle X_{e_0} X_{e_1},\ Z_{e_0} Z_{e_1},\ Z_1, \ldots, Z_{L-1} \rangle \ = \ \mathrm{stab}\big(\left|\Phi^+\right\rangle_{e_0 e_1} \otimes |0\rangle^{\otimes L-1} \big).$$

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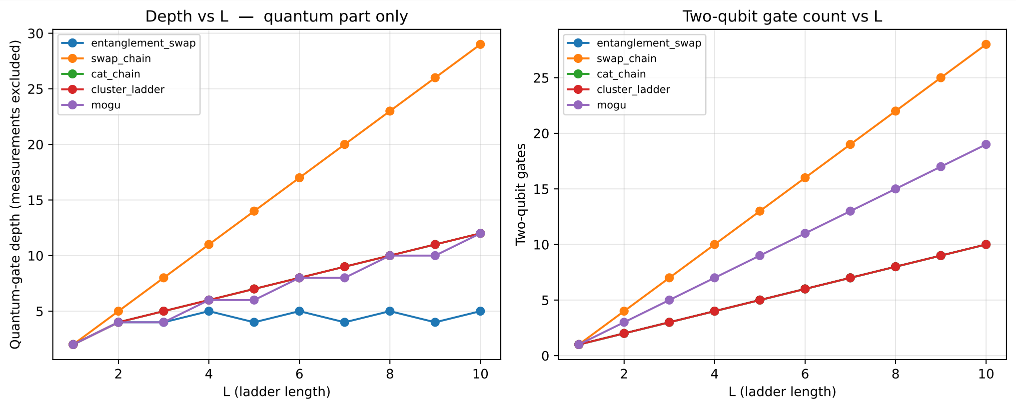
Parallel schedule (documentation-only, not in the verified code): if we interleave Phase C with the still-in-flight Phase B, the depth drops to

$$d_{\|}(L) \ \approx\ \begin{cases} \min(L+2,\ \lceil L/2\rceil+4) & L \text{ even}, \\ \min(L+1,\ \lceil L/2\rceil+3) & L \text{ odd}. \end{cases}$$

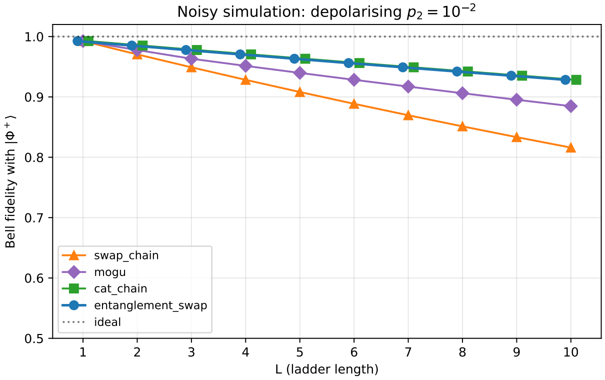
$\approx 2\times$ *depth reduction at large L — free constant-factor win.*

Benchmark: resources vs L



What you see. Protocol I (blue) is the only depth-$O(1)$ curve — its quantum part stays at ≤ 5 layers for every L. Measurements and feed-forward Paulis are excluded from this count (they execute in classical real time). MOGU is linear with the smallest slope among the unitary protocols; SWAP-chain dominates both axes.

Benchmark: Bell fidelity under depolarising noise



Markers for entanglement_swap and cat_chain are horizontally offset by ±0.1 for visibility; fidelities coincide at every integer L.

Symmetric two-qubit depolarising error $p_2 = 10^{-2}$ on every CNOT.

At $L = 10$**:**

- Protocol I: 0.928
- cat_chain: 0.928 (same 2Q-gate count)
- MOGU: 0.885
- swap_chain: 0.816

Protocol I (●) and cat_chain (■) give *identical* fidelities at every L — both incur $\approx N$ noisy CNOTs touching (e_0, e_1). The markers are jittered by ± 0.1 in L purely for visibility.

When to prefer which protocol

Hardware regime	Best choice	Why
Mid-circuit measurement + feed-forward	Protocol I	$O(1)$ depth
Unitary-only	MOGU	saturates LR bound
Small L, deep noise budget	cat_chain	fewest gates per endpoint

- Both protocols verified to fidelity $> 1 - 10^{-9}$ via `stim` Clifford and Qiskit `Statevector` for $L \leq 10$; spot checks at $L \in \{20, 30, 50\}$.
- **132 automated tests**, all passing.
- Single codebase under `analysis/`.

References & summary i

Take-home.

- Two protocols, one ladder, complementary hardware assumptions.
- Protocol I: $O(1)$ depth, classical feed-forward.
- Protocol II (MOGU): $L+2$ depth, measurement-free, depth-optimal for unitary 1D.

Key references.

- Żukowski, Zeilinger, Horne, Ekert, *Phys. Rev. Lett.* **71**, 4287 (1993) — entanglement swapping primitive.
- Briegel, Dür, Cirac, Zoller, *Phys. Rev. Lett.* **81**, 5932 (1998) — quantum repeater.
- Bäumer *et al.*, *PRX Quantum* **5**, 030339 (2024), arXiv:2308.13065 — long-range CNOT teleportation on 101 IBM qubits.
- Song *et al.*, *Phys. Rev. Applied* **24**, 024068 (2025), arXiv:2409.06989 — constant-depth fan-out with feedforward.

References & summary ii

Thank you.

Questions?